

Experiment 4

Module Name: Robot Framework

Experiment Name: Assertions and Verification

Problem Statement: Write a test case that performs an action and then uses an assertion to check whether the expected outcome matches the actual result. Implement a test case that verifies the response of an API request using the "Should Be Equal As Strings" keyword

Objective:. To perform strict validations on data states and API responses using Robot Framework assertions

Tools, Software, and Concepts Used:

Requests Library, BuiltIn Assertions

Source Code / Implementation Steps:

1. Create a folder inside that folder create 'Experiment4.robot' and write this:

```
*** Settings ***
Library      RequestsLibrary

*** Variables ***
${BASE_URL}  https://reqres.in

*** Test Cases ***
Verify API Response Code And Data
    [Documentation]    Sends a GET request and asserts the response status.
    Create Session      reqres    ${BASE_URL}
    ${response}=        GET On Session    reqres    /api/users/2

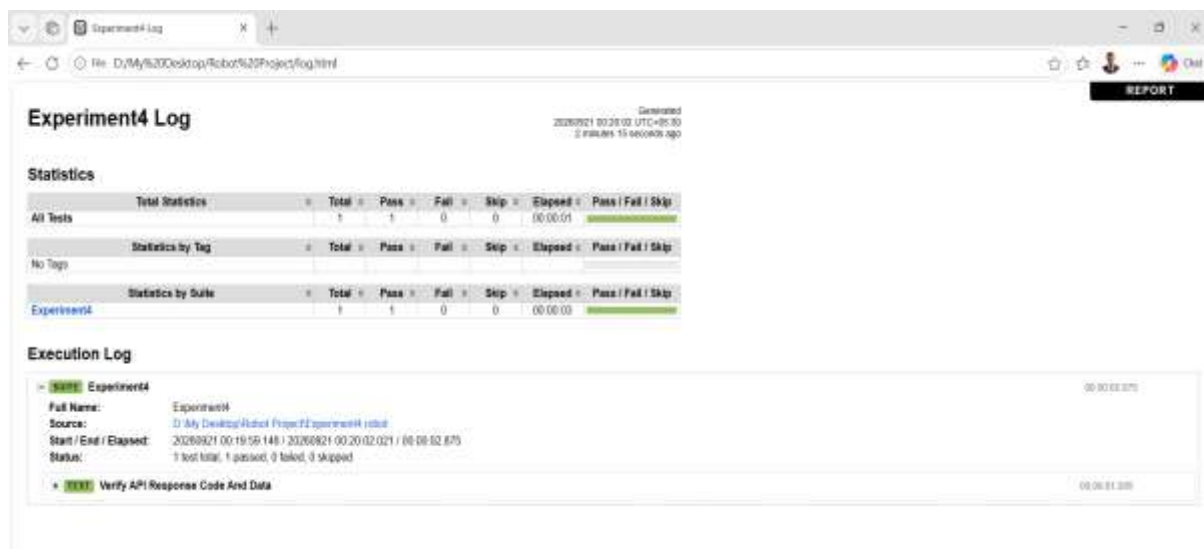
    # Assertion 1: Check status code matches 200
    ${status_code}=    Convert To String    ${response.status_code}
    Should Be Equal As Strings    ${status_code}    200

    # Assertion 2: Check standard built-in equality
    Should Be Equal As Strings    ${response.json()['data']['first_name']}
    Janet
```

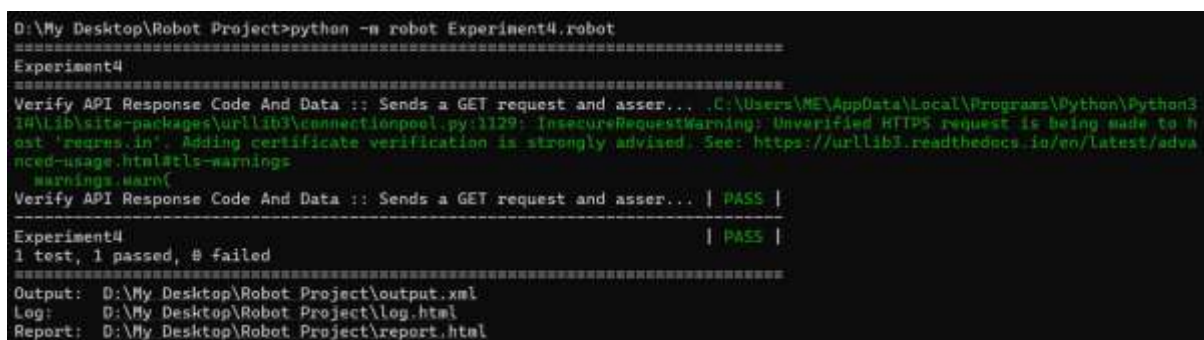
2. Now from the folder go to cmd and write this:

```
>>> python -m robot Experiment4.robot
```

Output with clearly labelled screenshots:



Screenshot1. Output of log.html



Screenshot 1: Terminal Output

Brief Result: The API GET request successfully returned the JSON payload, and both the status code and the `first_name` value passed the strict string assertions.

Observation: Converting non-string data types (like integer status codes) to strings is often necessary before using string-specific assertion keywords.

Conclusion: Assertions are the core of automated testing. Without them, a script only performs actions but does not validate functionality.